IPv6 Configuration and Subnetting

1. Assign and Verify a Static IPv6 Address

A static IPv6 address is manually configured and remains assigned until you change or remove it.

Windows steps

1. Open Command Prompt as Administrator.
2. Find your network adapter name:

netsh interface ipv6 show interfaces

3. Choose the active adapter, usually Wi-Fi or Ethernet.
4. Assign an example IPv6 address:

netsh interface ipv6 add address interface="Wi-Fi" address=2001:db8:1::10

2001:db8::/32 is reserved for documentation, so this address is only an example and should not be used for actual Internet connectivity. For a real static address, use an address assigned by your network administrator or a suitable local IPv6 prefix.

5. Verify the configuration:

ipconfig

Or:

netsh interface ipv6 show addresses

Linux alternative

ip -6 addr show

sudo ip -6 addr add 2001:db8:1::10/64 dev wlan0

ip -6 addr show dev wlan0

Replace wlan0 with your actual interface name, such as wlp2s0 or eth0. The example address is for documentation only.

Expected observation: The assigned address should appear in the adapter's IPv6 address list. The Linux ip addr add command generally configures the address temporarily; it does not automatically make it persistent after reboot.

2. Configure a Unique Local IPv6 Address (ULA)

IPv6 Unique Local Addresses begin with fd and are intended for private networks.

Example LAN addressing plan:

Device	**IPv6 Address**	**Prefix**
My laptop	fd12:3456:789a:1::10	/64

Friend's laptop	fd12:3456:789a:1::20	/64
Router LAN interface	fd12:3456:789a:1::1	/64

These are example addresses. Before using them, check that the prefix doesn't conflict with your existing LAN.

Windows commands

Open Command Prompt as Administrator.

netsh interface ipv6 add address interface="Wi-Fi" address=fd12:3456:789a:1::10

Check the address:

ipconfig

Then test the other device, assuming its address is fd12:3456:789a:1::20:

ping -6 fd12:3456:789a:1::20

Linux commands

sudo ip -6 addr add fd12:3456:789a:1::10/64 dev wlan0

ip -6 addr show dev wlan0

ping -6 fd12:3456:789a:1::20

Expected result: Successful replies indicate IPv6 connectivity between the devices. If ping fails, check that both devices are on the same LAN, the addresses and prefixes are correct, and the firewall permits ICMPv6 traffic.

3. Test a Temporary IPv6 Address

Temporary IPv6 addresses are often used to improve privacy by reducing the exposure of a device's stable interface identifier. Whether an address changes after reconnecting or rebooting depends on the operating system, privacy settings and network.

Windows

1. Open Command Prompt.
2. Run:

ipconfig

3. Record the IPv6 addresses shown for your Wi-Fi adapter.
4. Disconnect and reconnect Wi-Fi, then run ipconfig again.
5. Restart the computer and check again.

Linux

Check the addresses:

ip -6 addr show dev wlan0

Look for an address marked temporary or mngtmpaddr, depending on the configuration.

Reconnect to the network and check again:

ip -6 addr show dev wlan0

Record your observations

Test	IPv6 Address Observed	Did It Change?
Before reconnecting	Write your actual result	—
After reconnecting	Write your actual result	Yes/No
After rebooting	Write your actual result	Yes/No

Conclusion: A temporary IPv6 address may change over time or after reconnection, but this is not guaranteed. Record what actually happens on your system rather than assuming it changes.

4. Python Script: Search Restaurants by Name and IPv6 Address

This program stores restaurant names with their IPv6 addresses and lets the user search by typing part of a restaurant's name.

r = [("Zomato Cafe", "fd12:3456:789a:1::10"), ("Pizza Palace", "fd12:3456:789a:1::20"), ("Burger House", "fd12:3456:789a:1::30"), ("Spice Garden", "fd12:3456:789a:1::40")]s = input("Enter restaurant name: ").lower()f = Falsefor n, a in r: if s in n.lower(): print("Restaurant:", n) print("IPv6 Address:", a) f = Trueif not f: print("Restaurant not found")

Sample output

Enter restaurant name: pizza

Restaurant: Pizza Palace

IPv6 Address: fd12:3456:789a:1::20

Explanation:

- The list r stores each restaurant's name and IPv6 address.
- The program converts the search input to lowercase.
- It checks whether the entered text appears in each restaurant's name.
- If a match is found, the program displays the restaurant and its IPv6 address.

These addresses are sample values for demonstrating searching; they don't mean the restaurants actually operate on those IPv6 addresses.

5. IPv6 Subnetting Scenario with Three Subnets

Suppose a food delivery company uses the IPv6 prefix fd12:3456:789a::/48 for its internal network. We can allocate separate /64 subnets to three departments.

Department	IPv6 Subnet	Example Host Address
Delivery Partners	fd12:3456:789a:1::/64	fd12:3456:789a:1::10/64
Restaurants	fd12:3456:789a:2::/64	fd12:3456:789a:2::10/64
Administration	fd12:3456:789a:3::/64	fd12:3456:789a:3::10/64

Each subnet has a /64 prefix, which is the standard size for most IPv6 LANs.

Assign one subnet address to your system

For example, to assign the Delivery Partners subnet's sample address to a Windows Wi-Fi adapter:

netsh interface ipv6 add address interface="Wi-Fi" address=fd12:3456:789a:1::10

Verify it:

ipconfig

On Linux, use:

sudo ip -6 addr add fd12:3456:789a:1::10/64 dev wlan0

ip -6 addr show dev wlan0

Replace Wi-Fi or wlan0 with your actual adapter name. Only use the example prefix if it doesn't conflict with your existing network. To communicate with devices outside the local subnet, you also need suitable routing and firewall configuration.